Linear Regression and OLS

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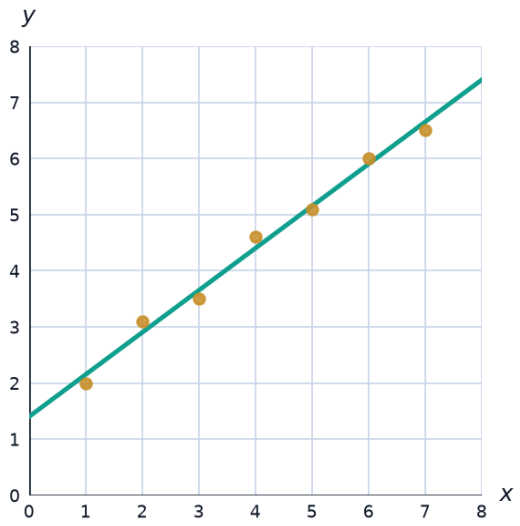
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A biostatistician is investigating the relationship between the dosage of a new experimental drug (measured in milligrams) and the reduction in systolic blood pressure (in mmHg) for patients with mild hypertension. They collected data from $n = 200$ patients and fit a simple linear regression model. The model's output yielded an estimated intercept of $\hat{\beta}_0 = 5.2$ and an estimated slope of $\hat{\beta}_1 = -0.8$. The standard error of the estimated slope is $\text{SE}(\hat{\beta}_1) = 0.15$. A 95% confidence interval for the true slope β_1 was calculated to be $[-1.12, -0.48]$. The biostatistician then constructs a hypothesis test to determine if the true slope is significantly different from zero. To do this, they calculate a test statistic. If the same α level used for the confidence interval were applied to a two-tailed hypothesis test of $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 eq 0$, what would be the approximate p-value associated with this test statistic? Show your complete algebraic work.

2 [Foundational] • Target: Simple Linear Regression and Residuals

Consider a simple linear regression model where the predicted value $\hat{y}$ is given by $\hat{y} = b_0 + b_1 x$. If we have two data points $(x_1, y_1) = (2, 5)$ and $(x_2, y_2) = (4, 9)$, and the Ordinary Least Squares (OLS) method yields the regression line $\hat{y} = 1 + 2x$, what is the predicted value $\hat{y}$ for $x = 3$? Show your complete algebraic work.



3 [Foundational] • Target: Analytical OLS Formulas and Covariance

Consider a simple linear regression model where the predicted value $\hat{y}$ is related to the independent variable x by the equation $\hat{y} = mx + c$. If the line of best fit passes through the points $(1, 5)$ and $(3, 11)$, what is the slope m of this line? Show your complete algebraic work.

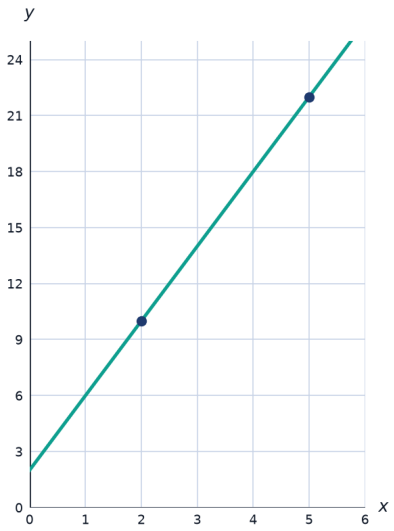
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4 [Foundational] • Target: Matrix Normal Equations and Projections

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$? Show your complete algebraic work.

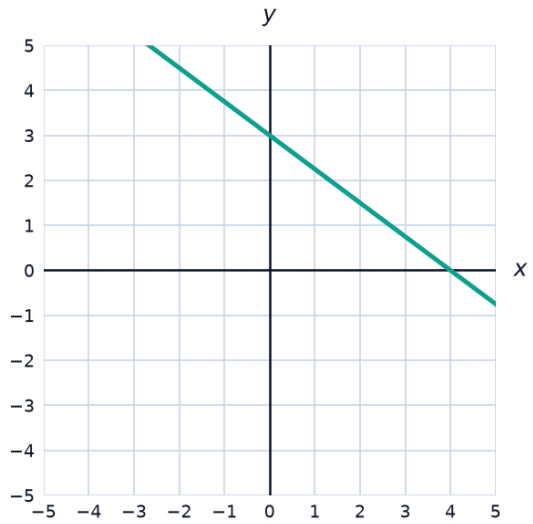
5 [Foundational] • Target: Goodness of Fit and Regularization

A simple linear regression model has been fitted to a dataset. The model equation is given by $\hat{y} = ax + b$. The regression line passes through the points $(2, 10)$ and $(5, 22)$ in the feature-target space. Determine the value of the intercept b.



6 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Consider a simple logistic regression model where the decision boundary is defined by the equation $3x + 4y - 12 = 0$. This equation represents a linear boundary in a 2D feature space. What is the y-intercept of this decision boundary? Show your complete algebraic work.



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7 [Foundational] • Target: Simple Linear Regression and Residuals

Consider a dataset of n observations (x_i, y_i) for $i = 1, \ldots, n$, where x_i are scalar predictor variables and y_i are scalar response variables. The ordinary least squares (OLS) estimates for the intercept β_0 and the slope β_1 of the linear model $y = \beta_0 + \beta_1 x + \varepsilon$ are given by $\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Suppose we have a dataset where $n = 5$, $\sum x_i = 15$, $\sum y_i = 30$, $\sum x_i^2 = 55$, $\sum y_i^2 = 210$, and $\sum x_i y_i = 105$. A new data point (x_6, y_6) is added to the dataset. If the new regression line that minimizes the sum of squared errors for the $n + 1 = 6$ points is parallel to the original regression line (i.e., $\hat{\beta}_1^{\text{new}} = \hat{\beta}_1^{\text{old}}$), what is the value of y_6 in terms of x_6? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Analytical OLS Formulas and Covariance

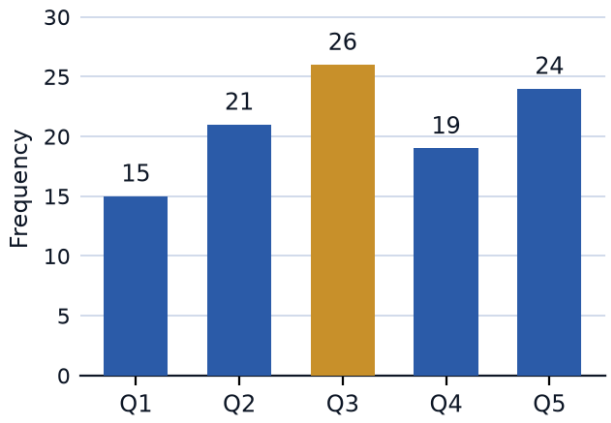
Consider the matrix $A = (1, 1, 0; 1, 0, 1; 0, 1, 1)$. Let $\mathbf{b} = (1; 1; 1)$. We want to find the vector $\hat{\mathbf{x}}$ that minimizes $Ax - b$. This is equivalent to finding the projection of $\mathbf{b}$ onto the column space of A, denoted by $\text{Col}(A)$. The normal equations for this least squares problem are $A^T A\mathbf{x} = A^T\mathbf{b}$. The unique solution $\hat{\mathbf{x}}$ to the normal equations yields the best approximate solution. Calculate the value of $\hat{\mathbf{x}}_1$, the first component of $\hat{\mathbf{x}}$.

9 [Intermediate] • Target: Matrix Normal Equations and Projections

Consider a 2D dataset represented by the points $A = (1, 2)$, $B = (4, 2)$, and $C = (2, 5)$. If this dataset undergoes a linear transformation represented by the matrix $M = (2, 0; 0, 1)$, what are the coordinates of the transformed points A', B', and C'? Show your complete algebraic work.

10 [Intermediate] • Target: Goodness of Fit and Regularization

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x? Show your complete algebraic work.



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11 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider a simple logistic regression model where the decision boundary is defined by the linear equation $w_1x_1 + w_2x_2 + b = 0$. If the weights are $w_1 = 2$ and $w_2 = 3$, and the bias is $b = -7$, what is the equation of the decision boundary? Show your complete algebraic work.

12 [Intermediate] • Target: Simple Linear Regression and Residuals

Consider a simple linear regression model where a dependent variable y is predicted from an independent variable x. The model is given by the equation $y = \beta_0 + \beta_1 x + \varepsilon$, where ε is the error term. Suppose we have estimated the regression line to be $\hat{y} = 3 + 2x$. If we observe an independent variable value of $x = 4$, what is the predicted value $\hat{y}$ for the dependent variable? Show your complete algebraic work.

13 [Intermediate] • Target: Analytical OLS Formulas and Covariance

Consider the following system of linear equations represented by an augmented matrix: $[1, 2, |, 5; 0, 1, |, 2]$. What is the value of x in the first equation? Show your complete algebraic work.

14 [Intermediate] • Target: Matrix Normal Equations and Projections

Consider the system of linear equations represented by the following augmented matrix: $(1, 2, |, 5; 0, 1, |, 2)$ Use row reduction to determine the unique solution for x and y in the system. What is the value of x? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

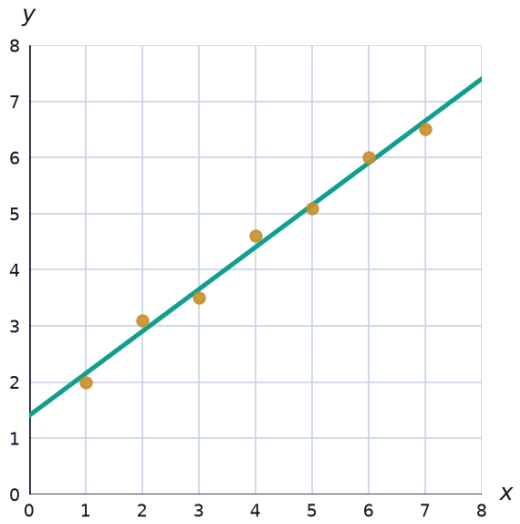
15 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 7.01 Logistic Regression and Classification

For a bivariate feature dataset, $\text{Cov}(x, y) = 36$, $\text{Var}(x) = 12$, $\bar{x} = 5$, and $\bar{y} = 35$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

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16 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

For a bivariate feature dataset, $\text{Cov}(x, y) = 48$, $\text{Var}(x) = 16$, $\bar{x} = 5$, and $\bar{y} = 40$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.



17 [Challenging] • Target: Simple Linear Regression and Residuals • Cross-Lesson: 1.01 Vectors and Embeddings in AI

For a bivariate feature dataset, $\text{Cov}(x, y) = 60$, $\text{Var}(x) = 20$, $\bar{x} = 5$, and $\bar{y} = 45$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

18 [Challenging] • Target: Analytical OLS Formulas and Covariance • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

For a bivariate feature dataset, $\text{Cov}(x, y) = 72$, $\text{Var}(x) = 24$, $\bar{x} = 5$, and $\bar{y} = 50$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

19 [Challenging] • Target: Matrix Normal Equations and Projections • Cross-Lesson: 3.01 Gradient Descent and Optimization

For a bivariate feature dataset, $\text{Cov}(x, y) = 12$, $\text{Var}(x) = 4$, $\bar{x} = 5$, and $\bar{y} = 25$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

20 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

For a bivariate feature dataset, $\text{Cov}(x, y) = 24$, $\text{Var}(x) = 8$, $\bar{x} = 5$, and $\bar{y} = 30$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 2x + 3$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. 11

b. 14

c. 7

d. 8

22 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 3x + 6$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. 18

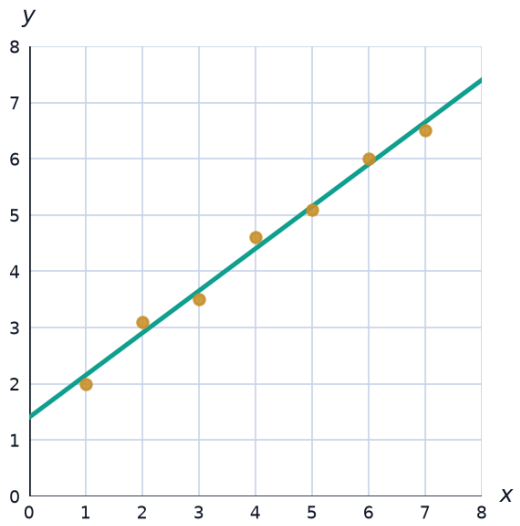
b. 21

c. 14

d. 12

23 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 4x + 9$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?



a. 25

b. 28

c. 21

d. 16

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24 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 5x + 12$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. 32

b. 35

c. 28

d. 20

25 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 6x + 15$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. 39

b. 42

c. 35

d. 24

26 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. 46

b. 49

c. 42

d. 28

27 [Foundational] • Target: Linear Regression & OLS Practice Exercises

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 20$ and the Total Sum of Squares is $\text{SST} = 100$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$

b. $R^2 = 0.20$

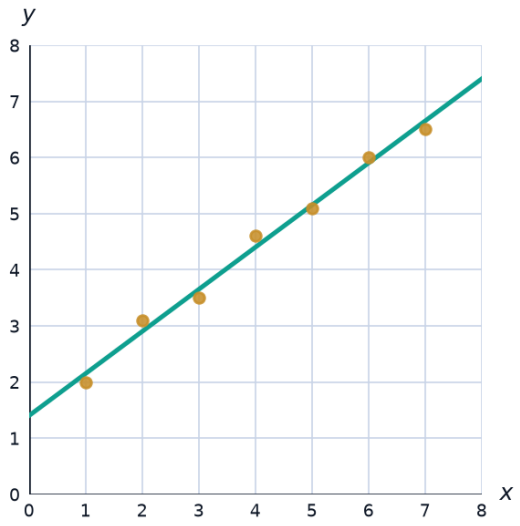
c. $R^2 = 0.90$

d. $R^2 = 0.25$

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28 [Foundational] • Target: Linear Regression & OLS Practice Exercises

In a multiple linear regression model, the Sum of Squared Errors is \(\text{SSE} = 40 \) and the Total Sum of Squares is \(\text{SST} = 200 \). What is the coefficient of determination \(R^2 \)?



a. \(R^2 = 0.80 \)

b. \(R^2 = 0.20 \)

c. \(R^2 = 0.90 \)

d. \(R^2 = 0.25 \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Simple Linear Regression and Residuals

In simple linear regression $\hat{y} = 3.5x + 12$, what is the residual $e_i = y_i - \hat{y}_i$ for an observation at $(x_i, y_i) = (4, 30)$?

a. $+4$

b. -4

c. $+2$

d. -2

30 [Intermediate] • Target: Simple Linear Regression and Residuals

What is the mathematical condition guaranteed by the OLS fit regarding the sum of all residuals $\sum_{i=1}^{n} e_i$?

a. $\sum e_i = 0$ exactly

b. $\sum e_i = 1$

c. $\sum e_i = n$

d. $\sum e_i = \text{SSE}$

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31 [Intermediate] • Target: Simple Linear Regression and Residuals

Given $\bar{x} = 10$, $\bar{y} = 25$, and fitted slope $\hat{\beta}_1 = 2$, compute the y-intercept $\hat{\beta}_0$.

a. 5

b. 15

c. 20

d. 10

32 [Intermediate] • Target: Analytical OLS Formulas and Covariance

For a dataset with $\text{Cov}(x, y) = 18$ and $\text{Var}(x) = 6$, compute the OLS slope $\hat{\beta}_1$.

a. 3.0

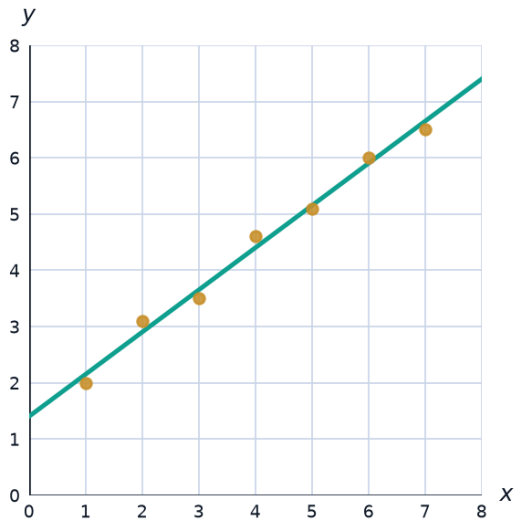
b. 0.33

c. 12

d. 108

33 [Intermediate] • Target: Analytical OLS Formulas and Covariance

If the Pearson correlation coefficient between x and y is $r = 0.8$, with $\sigma_x = 2$ and $\sigma_y = 5$, find the slope $\hat{\beta}_1$.



a. 2.0

b. 1.25

c. 0.32

d. 4.0

34 [Intermediate] • Target: Analytical OLS Formulas and Covariance

If all x values in a dataset are scaled by a factor of 10, how does the OLS regression slope change?

a. It is divided by 10

b. It is multiplied by 10

c. It is multiplied by 100

d. It remains unchanged

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35 [Intermediate] • Target: Matrix Normal Equations and Projections

State the matrix Normal Equations for the OLS weight vector $\boldsymbol{\beta}$ given design matrix $\mathbf{X}$ and target $\mathbf{y}$.

a. $(\mathbf{X}^T\mathbf{X})\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$

b. $\mathbf{X}\boldsymbol{\beta} = \mathbf{y}$

c. $(\mathbf{X}\mathbf{X}^T)\boldsymbol{\beta} = \mathbf{y}$

d. $\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$

36 [Intermediate] • Target: Matrix Normal Equations and Projections

When is the Gram matrix $(\mathbf{X}^T\mathbf{X})$ strictly invertible?

a. When the columns of $\mathbf{X}$ are linearly independent (full column rank)

b. When $\mathbf{X}$ is a square matrix

c. When all elements in $\mathbf{X}$ are positive

d. When the target $\mathbf{y}$ has zero mean

37 [Intermediate] • Target: Matrix Normal Equations and Projections

What geometric role does the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ play?

a. It orthogonal-projects target $\mathbf{y}$ onto the column space of $\mathbf{X}$ ($\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$)

b. It inverts the covariance matrix

c. It computes feature cross-correlations

d. It normalizes features to zero mean

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

In an evaluation, $\text{SSE} = 25$ and $\text{SST} = 100$. Compute the coefficient of determination R^2.

a. 0.75

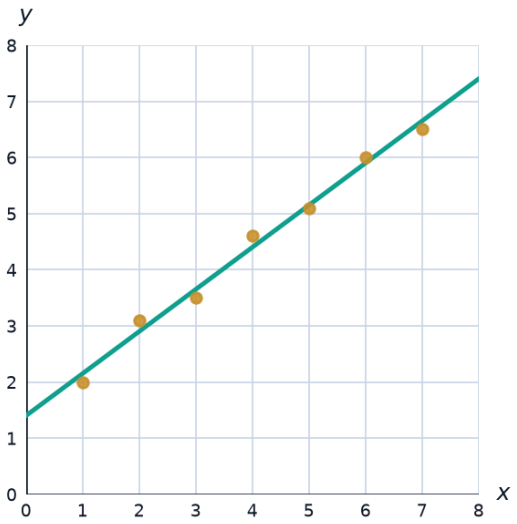
b. 0.25

c. 0.50

d. 0.80

39 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 3.01 Gradient Descent and Optimization

How does Ridge regression (L_2 penalty) resolve multicollinearity when $\mathbf{X}^T\mathbf{X}$ is ill-conditioned?



a. It adds $\lambda\mathbf{I}$ to $\mathbf{X}^T\mathbf{X}$, guaranteeing invertibility: $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}$

b. It drops half the features automatically

c. It sets all correlated weights to zero

d. It eliminates the bias term

40 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

If $R^2 = 0.64$ in simple linear regression, what is the magnitude of the Pearson correlation coefficient $|r|$?

a. 0.80

b. 0.64

c. 0.32

d. 0.40

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41 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 100$ and the Total Sum of Squares is $\text{SST} = 500$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

42 [Challenging] • Target: Simple Linear Regression and Residuals • Cross-Lesson: 7.01 Logistic Regression and Classification

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 120$ and the Total Sum of Squares is $\text{SST} = 600$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

43 [Challenging] • Target: Analytical OLS Formulas and Covariance • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 20$ and the Total Sum of Squares is $\text{SST} = 100$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

44 [Challenging] • Target: Matrix Normal Equations and Projections • Cross-Lesson: 1.01 Vectors and Embeddings in AI

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 40$ and the Total Sum of Squares is $\text{SST} = 200$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

45 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 60$ and the Total Sum of Squares is $\text{SST} = 300$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

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